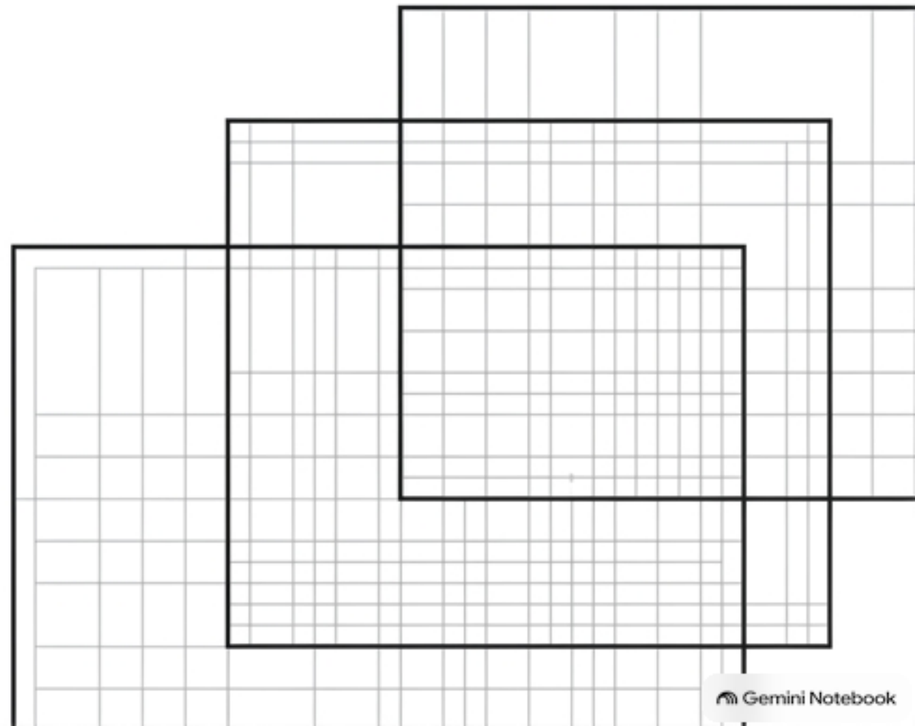
MODULE 11, LESSON 3

AI 101 for Mechanical Engineers

BATCHES, EPOCHS AND REALISTIC EXPECTATIONS



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Lesson Outline

1.	Scope and Module Limitations
2.	Fundamental Computational Parameters
3.	Training Dynamics and Generalization
4.	Parameter Application and Assessment

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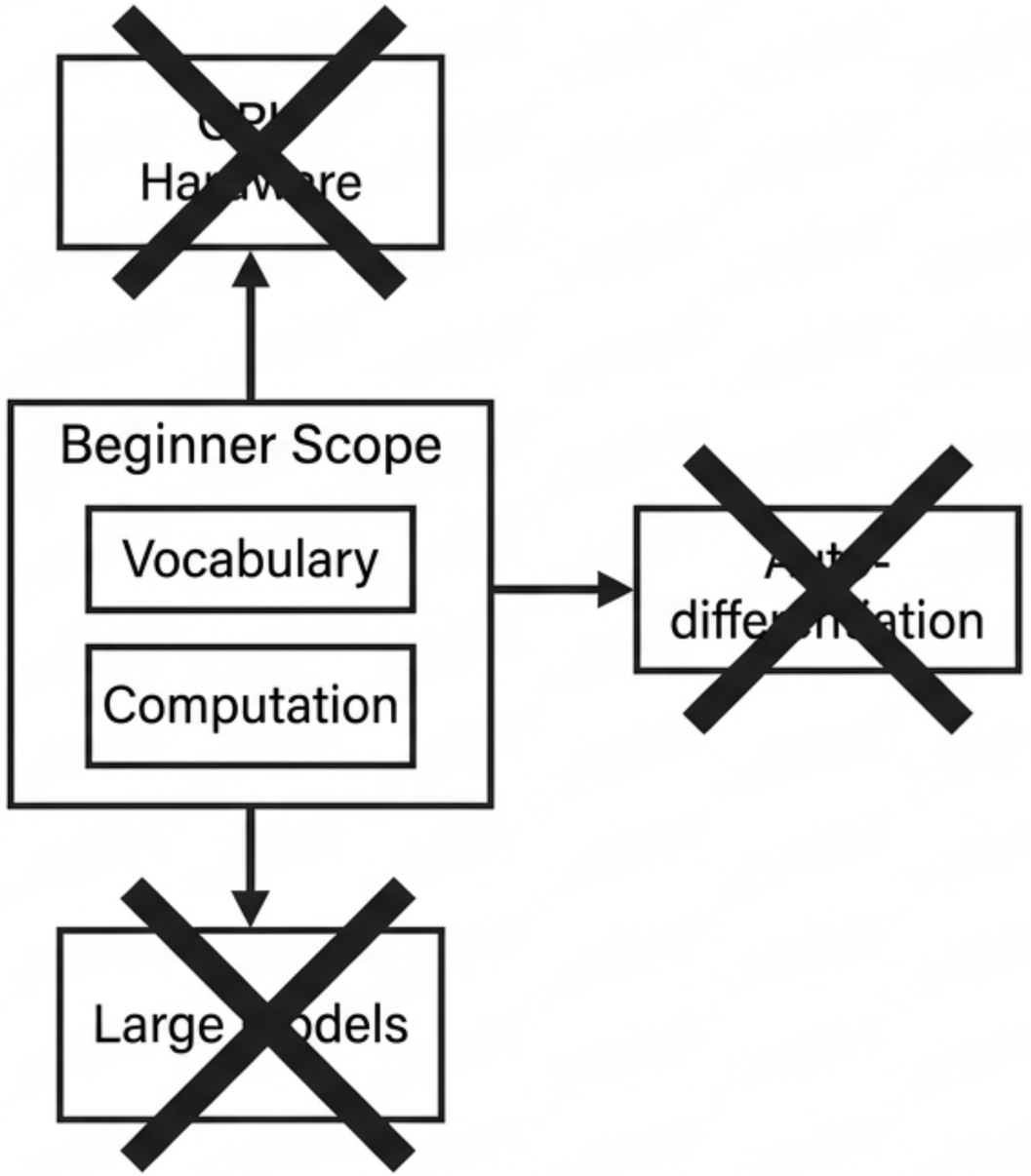
1. Scope and Module Limitations

Defines the computational and architectural boundaries for this beginner-level module.

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Module Computational Boundaries

- Primary objective is understanding core computational vocabulary.
- Hardware constraints: Graphics Processing Units are not required.
- Excludes comprehensive neural network architecture design procedures.
- Excludes automatic differentiation software utilization.
- Excludes large-scale model training methodologies.



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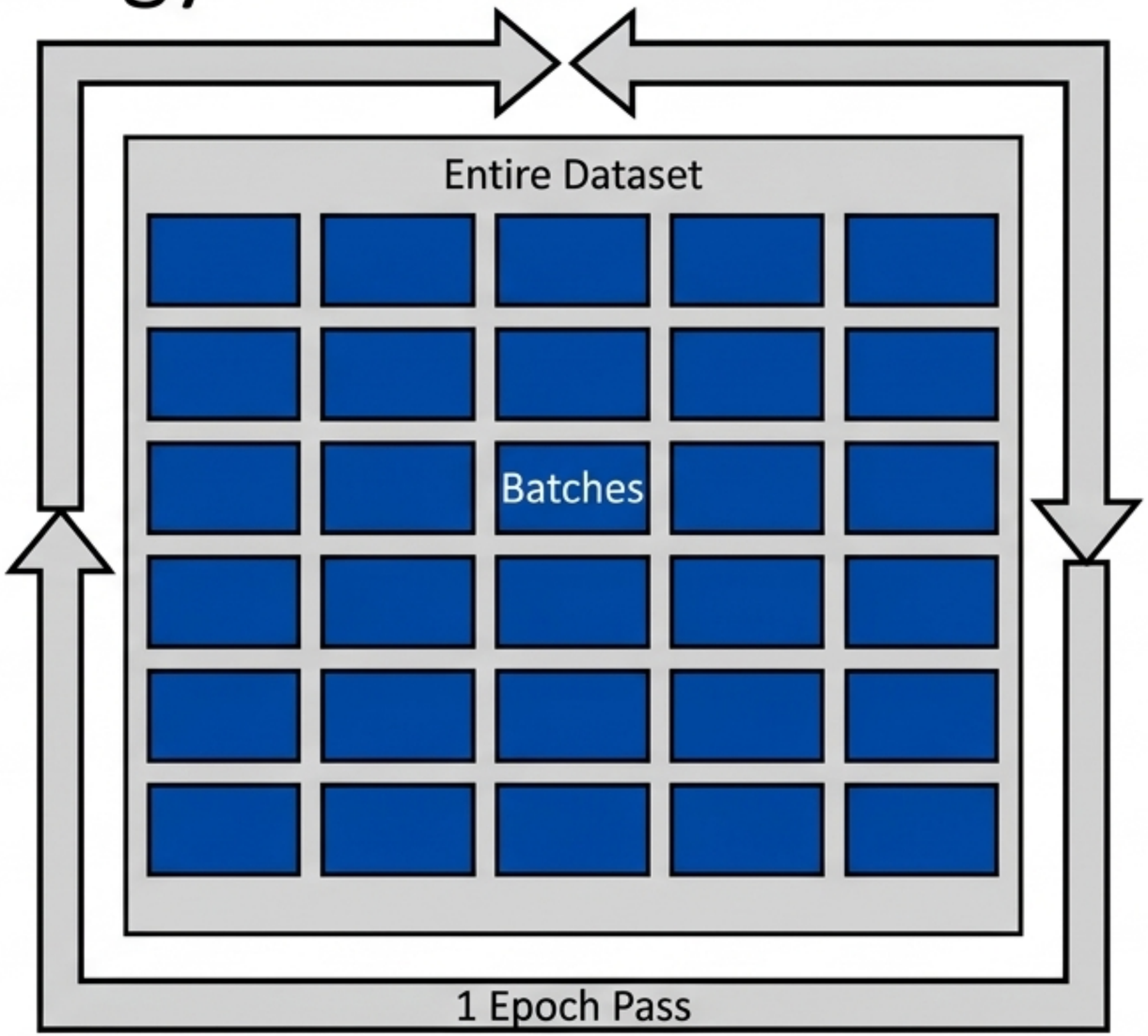
2. Fundamental Computational Parameters

Establishes the definitions and quantitative relationships governing dataset processing.

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Data Partitioning Terminology

- Epoch: One complete pass through the entire training dataset.
- Batch: Discrete subset of training examples processed per update.
- Mini-batch procedure: Standard methodology utilizing segmented example groups.
- Indivisible subsets require explicitly stated algorithmic treatment rules.



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Standard Update Calculation

- Updates dictate model adjustment frequency per epoch pass.
- Quantity of standard updates derives from total dataset proportions.
- Example computation yields twenty standard algorithmic updates per epoch.

$$U = \frac{N_{total}}{N_{batch}}$$

U: Number of standard updates [-]
N_{total}: Total training samples [-] (Example: 1000)
N_{batch}: Samples per batch [-] (Example: 50)

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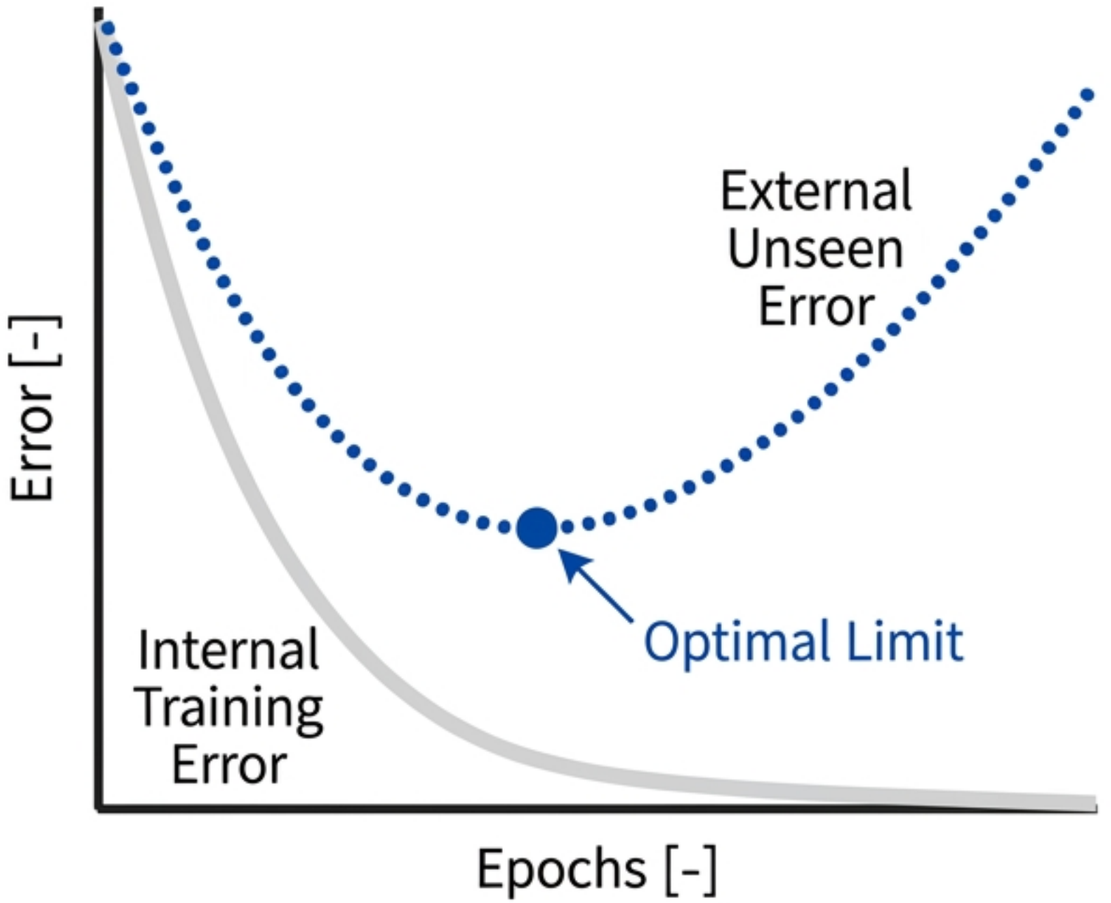
3. Training Dynamics and Generalization

Analyzes the relationship between extended training duration and external performance validity.

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Limitations of Additional Epochs

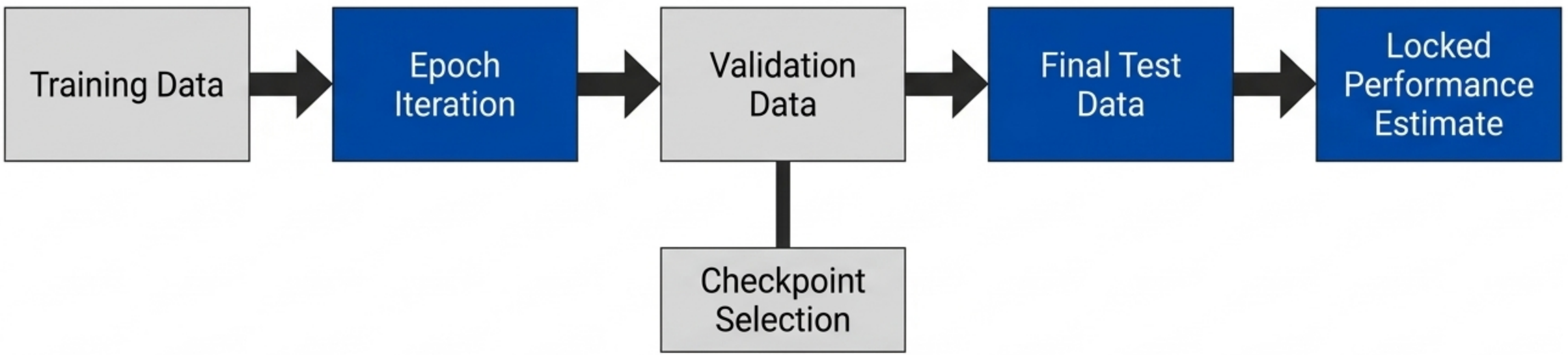
- Deep networks successfully formulate representations for highly complex data.
- Increased epoch counts reliably decrease internal training error metrics.
- Additional epochs guarantee zero improvement regarding unseen external performance.
- Excessive training iterations fit algorithms to training-specific variations.



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Performance Verification Sequence

- Validation datasets govern the selection of optimal model checkpoints.
- Validation procedures monitor operational boundaries and verify physical limits.
- Locked final test datasets estimate post-training operational performance.



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4. Parameter Application and Assessment

Consolidates definitions and validates comprehension of data processing mechanics.

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Diagnostic Parameter Summary

- Synthetic numerical evidence cannot substitute for physical equipment validation.
- Systematic verification ensures algorithms respect defined engineering constraints.

Parameter	Definition	Expected System Impact
Batch	Subset of examples for one algorithmic update	Governs update frequency
Epoch	Singular pass through entire training dataset	Decreases internal training error
Additional Epochs	Supplementary dataset iterations	No guaranteed external performance gain

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Key Points

- Scope restricts to fundamental computational vocabulary without advanced hardware.
- Epochs signify full dataset passes; batches dictate update quantities.
- Extended training risks fitting variation without improving external generalization.
- Validation datasets select checkpoints; isolated testing confirms final performance.

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